Female Genital Tract Pathology
LG-2
Diseases Of Vulva, Vagina & Cervix

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Lecture Outline:

- Diseases of vulva
- Diseases of vagina
- Diseases of cervix

Vulvitis: Inflammatory and infectious conditions

- Noninfectious vulvitis: irritant or allergic contact dermatitis; clinical erythema, pruritus and discomfort.
- Infectious vulvitis : Candida species, HSV, HPV-related lesions, and sexually transmitted bacterial infections.
- Histology varies by cause; ranges from superficial inflammations to squamous intraepithelial lesions.
- Persistent vulvar pruritus or abnormal white plaques require evaluation for chronic dermatoses or neoplasia.

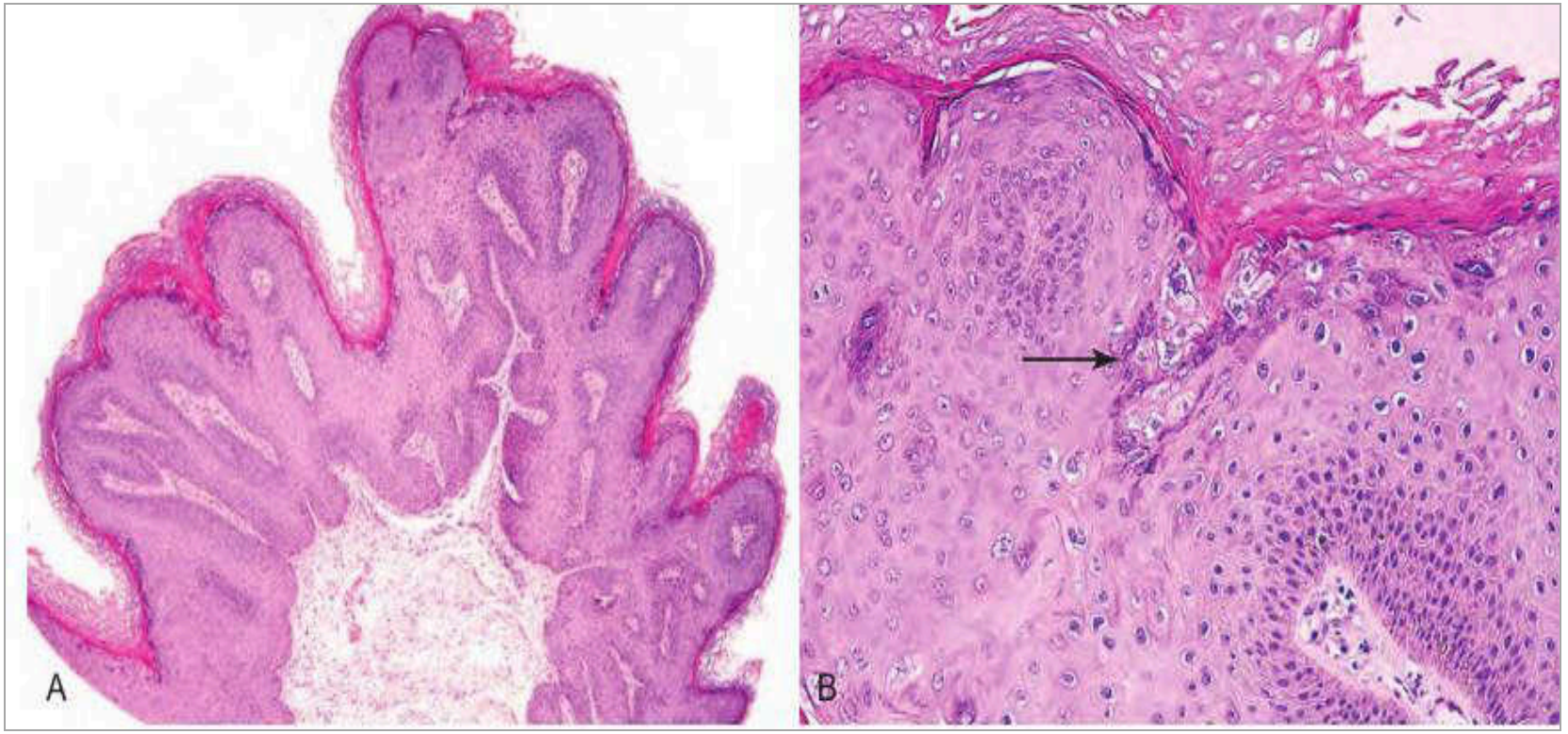
Bartholin duct cyst and abscess

- Obstruction of a Bartholin gland duct produces retention of secretions and a duct cyst.
- Secondary infection may cause a painful abscess however; the cyst itself is often painless.

Tumors of vulva:

Condyloma acuminatum (genital wart)

- Benign exophytic squamous proliferation most often associated with low-risk HPV 6 and 11.
- Transmission is by sexual contact
- Gross appearance: soft, papillary or verrucous lesion.
- Microscopy: fibrovascular papillae, acanthosis, hyperkeratosis and koilocytosis.
- Koilocytes show perinuclear clearing and enlarged, irregular, hyperchromatic nuclei.



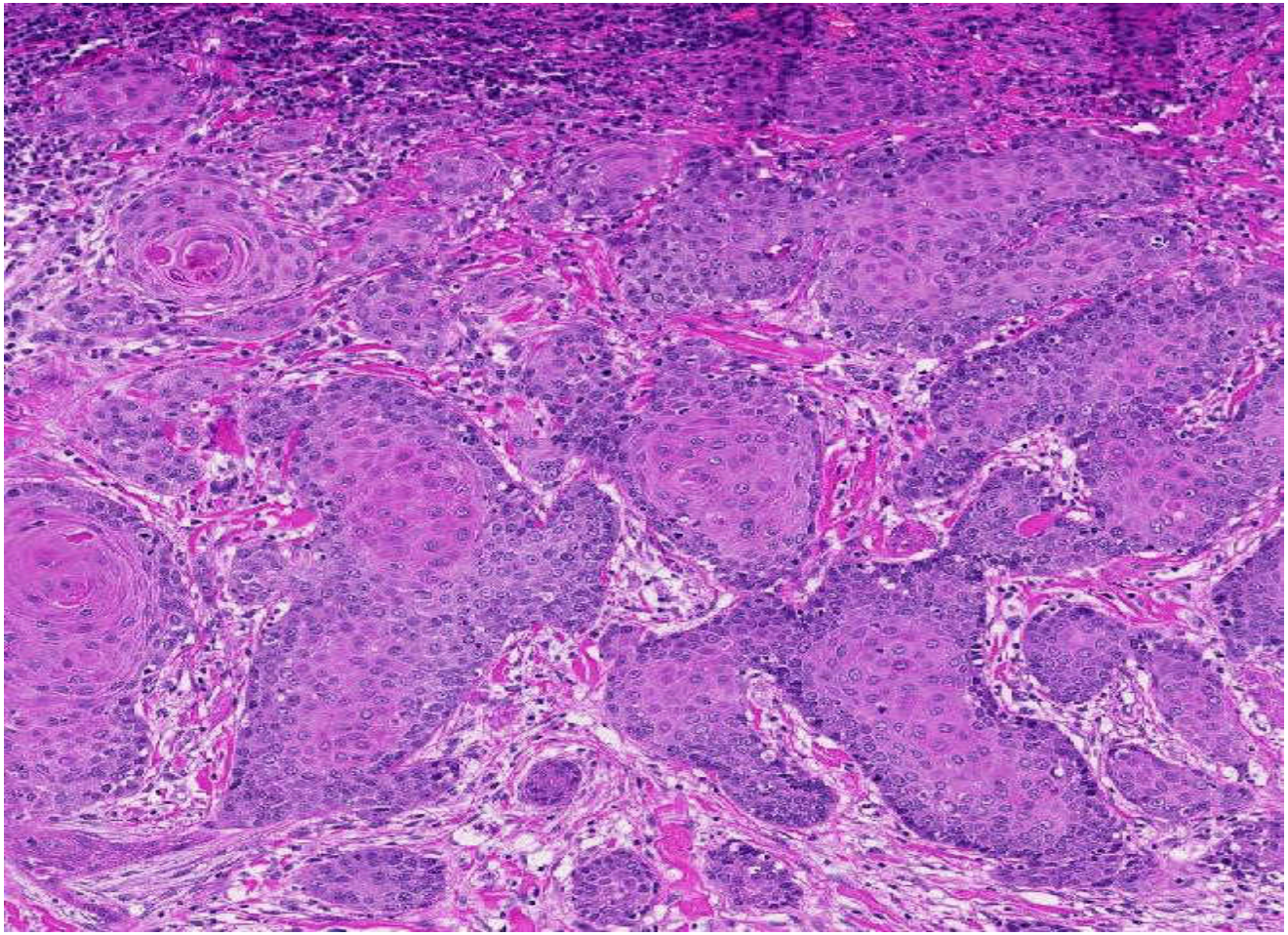
•Condyloma acuminatum. A, Low-power view showing exophytic, papillary architecture.
B, High-power view reveals HPV cytopathic effect (koilocytic atypia) characterized by atypical, enlarged, hyperchromatic nuclei with perinuclear halos (arrow).

Carcinoma of vulva:

- 90% of carcinomas are squamous cell carcinomas
- Other tumors are mainly adenocarcinomas or basal cell carcinomas

Vulvar squamous cell carcinoma

- Squamous cell carcinoma accounts for the great majority of primary vulvar malignancies.
- HPV-associated pathway: usually high-risk HPV (particularly HPV 16), arise through HPV associated vulvar intraepithelial neoplasia (VIN).
- HPV-independent pathway: often associated with lichen sclerosus and HPV independent vulvar intraepithelial neoplasia.
- Gross: ulcerated, nodular or exophytic lesion; histology shows invasive squamous nests.



Vulval invasive squamous cell carcinoma

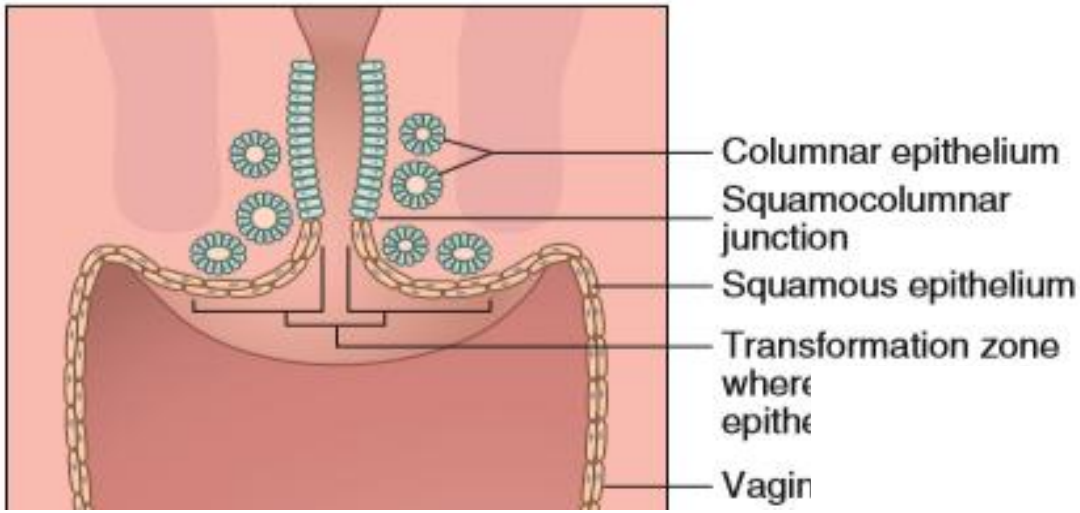
Vaginitis:

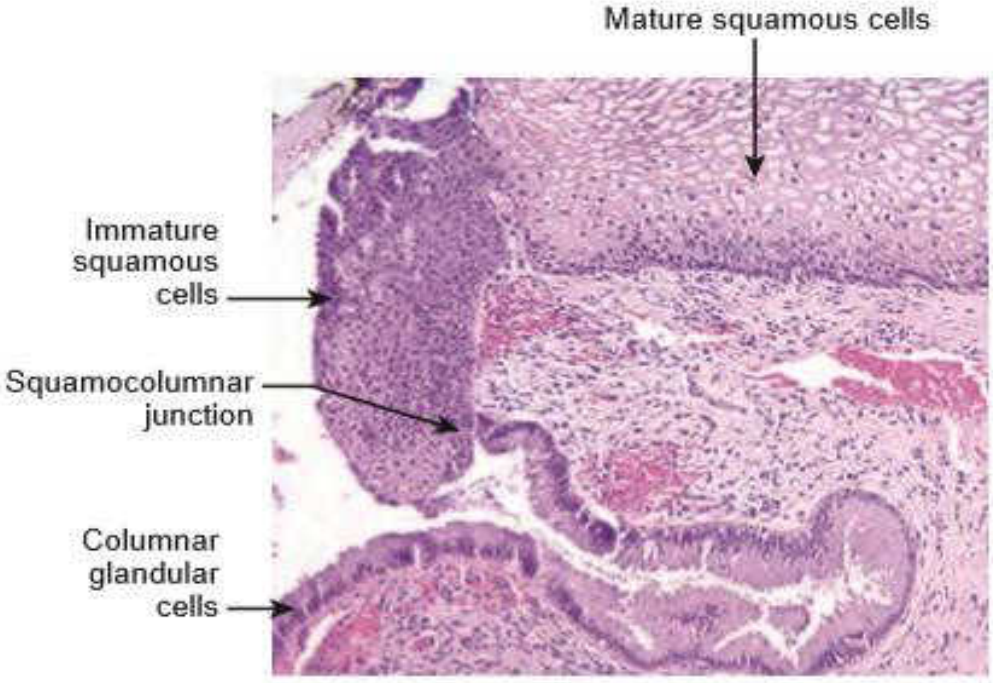
- Vaginitis is a relatively common condition that is usually transient and of no clinical consequence.
- A large variety of organisms have been implicated, including bacteria, fungi, and parasites.
- Frequent offenders, are Candida albicans and Trichomonas vaginalis.

Vaginal carcinoma:

- Primary vaginal carcinoma is rare; its usually extension or metastasis from cervix or vulva.
- Most primary vaginal cancers are squamous cell carcinomas, often associated with high-risk HPV.
- Vaginal HSIL (VaIN) is an important precursor lesion.
- Typically presents in older adults with vaginal bleeding or a visible lesion.

Cervix


Columnar epithelium
Squamocolumnar junction
Squamous epithelium
Transformation zone where epithe
Vagin


Mature squamous cells
Immature squamous cells
Squamocolumnar junction
Columnar glandular cells

Cervicitis:

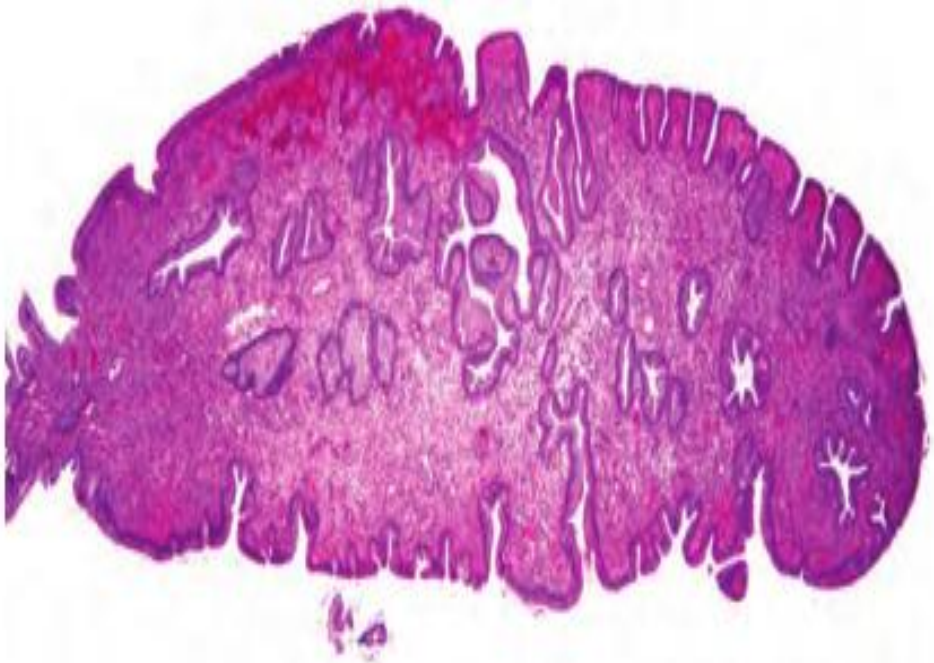
Cervicitis can be subclassified as:

- Infectious (specific) or non-infectious (non-specific)
- Acute or chronic

- Acute cervicitis: neutrophils seen in endocervical epithelium and stroma;
- Chronic cervicitis: lymphocytes and plasma cells are seen; often nonspecific.
- Important infectious causes include Chlamydia trachomatis and Neisseria gonorrhoeae; HSV and Trichomonas may also involve cervix.

Endocervical polyp

- Are benign polypoid proliferations of endocervical mucosa seen protruding from the endocervical mucosa (some times from external os).
- Microscopy: mucinous endocervical glands in edematous or fibrous, vascular stroma, often inflamed.
- May cause intermenstrual or postcoital spotting; many are asymptomatic.





Endocervical polyp composed of a dense fibrous stroma covered with endocervical columnar epithelium, Note the mucous secreting cervical epithelium.

Cervical squamous intraepithelial lesions

- Persistent infection by oncogenic HPV, particularly types 16 and 18, underlies most cervical carcinomas.
- HPV enters immature squamous cells, especially in the transformation zone.
- LSIL generally corresponds to CIN 1.
- HSIL generally corresponds to CIN 2–3 and carries greater progression risk.
- An intraepithelial lesion has an intact basement membrane: invasion defines carcinoma.

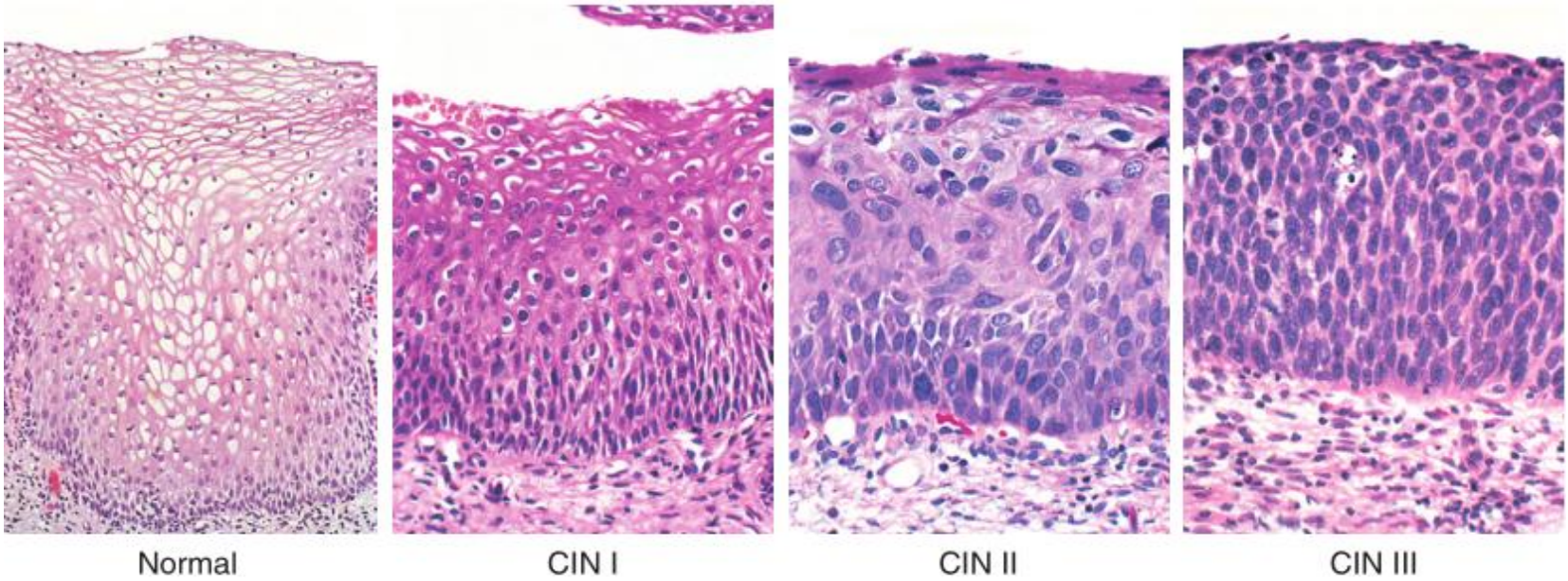
Classification of CIN:

Dysplasia/Carcinoma in Situ	Cervical Intraepithelial Neoplasia (CIN)	Squamous Intraepithelial Lesion (SIL), Current Classification
Mild dysplasia	CIN I	Low-grade SIL (LSIL)
Moderate dysplasia	CIN II	High-grade SIL (HSIL)
Severe dysplasia	CIN III	High-grade SIL (HSIL)
Carcinoma in situ	CIN III	High-grade SIL (HSIL)

CIN, Cervical intraepithelial neoplasia; SIL, squamous intraepithelial lesion.

Histologic grading: CIN and SIL

- CIN 1 / LSIL: atypia mainly in basal one-third; maturation preserved; koilocytosis may occur.
- CIN 2 / HSIL: atypical immature cells extend into middle epithelial layers; mitoses may occur above basal layer.
- CIN 3 / HSIL: severe atypia and loss of maturation through most or all epithelial thickness.



Spectrum of cervical intraepithelial neoplasia (CIN), with normal squamous epithelium for comparison: CIN I with atypia confined basal layer; CIN II with progressive atypia in two third of epithelium; and CIN III (carcinoma in situ) with diffuse (full thickness) atypia and loss of maturation.

Cervical cancer screening and prevention

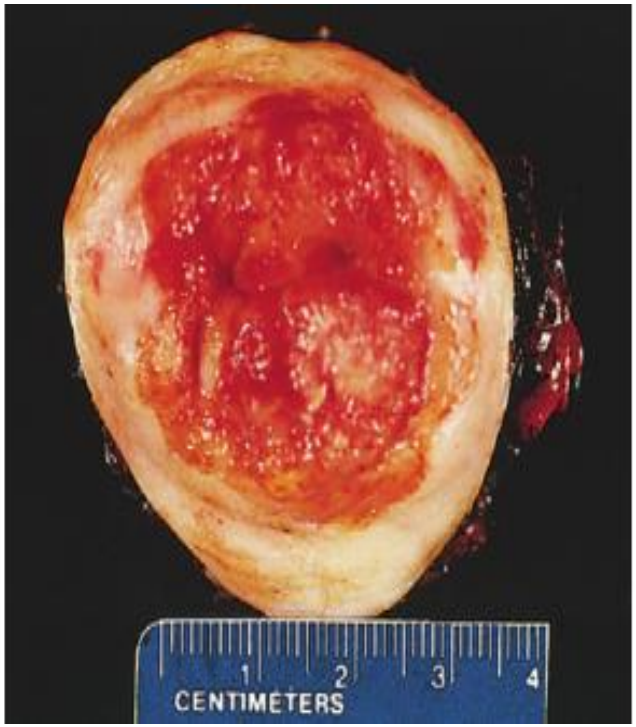
- Cervical cytology (Pap testing) detects abnormal exfoliated squamous cells.
- High-risk HPV testing detects oncogenic viral types.
- Abnormal screening results may require colposcopy with biopsy for histologic diagnosis.
- Screening recommendations vary with age, risk and country.
- Population-based HPV vaccination and screening reduce cervical cancer burden.

Cervical carcinoma: histologic types

- **Squamous cell carcinoma** is the most common histologic subtype of cervical cancer, accounting for approximately 80% of cases.
- Followed by **adenocarcinoma** (15%)
- **Adenosquamous and neuroendocrine** carcinomas are rare cervical tumors that account for the remaining 5% of cases.

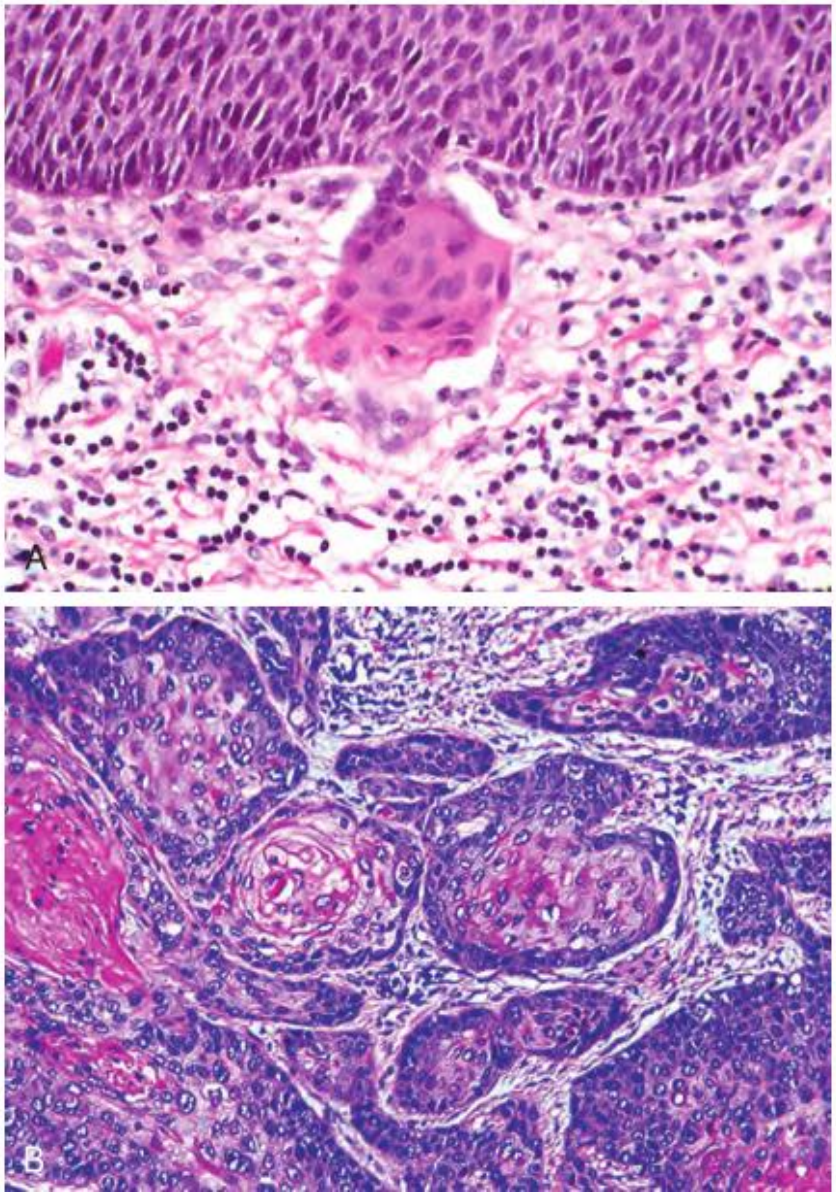
Cervical carcinoma:

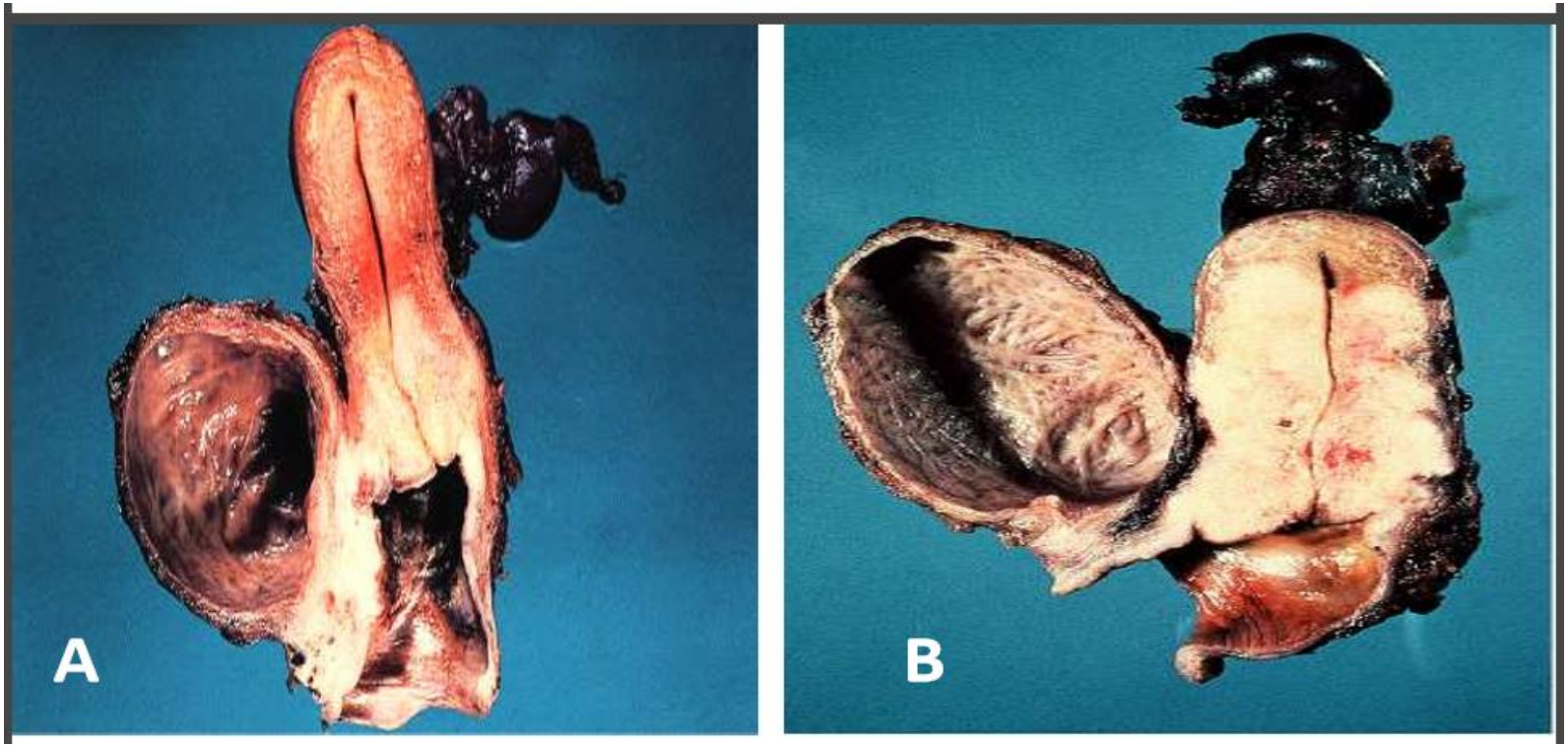
- Persistent high-risk HPV infection is the major driver of cervical squamous carcinoma and many adenocarcinomas.
- Viral E6 promotes degradation of p53; E7 functionally inactivates RB.
- The peak incidence of invasive cervical carcinoma is 45 years.



Cervical os with surrounding, invasive, exophytic cervical carcinoma

Squamous cell carcinoma of the cervix.
A, Microinvasive squamous cell carcinoma with invasive nest breaking through the basement membrane of high grade squamous intraepithelial lesion.
B, Invasive squamous cell carcinoma.





Squamous cell carcinoma of cervix :

A: extension into bladder.

B: massive extension into uterine corpus.

Clinical behavior and prognosis

- Early lesions may be asymptomatic; abnormal vaginal bleeding, especially postcoital bleeding, is a common presentation.
- Local extension may involve vagina, parametria, bladder or rectum; lymphatic spread involves pelvic nodes.
- Stage at diagnosis is the strongest major prognostic factor; nodal disease and histologic type also matter.
- Small-cell neuroendocrine carcinoma is uncommon and typically clinically aggressive.
- Most patients with stage IV cancer die as a consequence of local extension of the tumor rather than distant metastases.

References:

- Kumar, Abbas & Aster. Robbins Basic Pathology, 10th ed. Female Genital Tract chapter.
- Kumar, Abbas & Aster. Robbins & Cotran Pathologic Basis of Disease, 10th ed.

Thank you